

ASSIGNMENT 2.4

Q1. Connect a USB webcam to your computer, install any required drivers, and verify that the webcam works by opening the default camera app and capturing a photo.

➤ USB Webcam Setup and Testing

1. **Connect the webcam** – Plug the USB webcam into an available USB port on the computer.
2. **Install drivers** – Wait for Windows to automatically install the required drivers. If needed, install the driver provided by the webcam manufacturer.
3. **Check the webcam** – Open Device Manager → Cameras and confirm that the webcam is listed without any warning icon.
4. **Open Camera app** – Open the Windows Camera app from the Start menu.
5. **Select the webcam** – Make sure the external USB webcam is selected as the camera source.
6. **Capture a photo** – Click the Take Photo button and verify that the image is captured successfully.
7. **Confirm working** – If the webcam displays a clear live preview and the photo is saved, the webcam is working correctly.

Q2. Set up a local printer (USB or network), install the correct driver, and print a test page to confirm successful installation.

Step 1: Connect the Printer

- **For USB:** Plug the printer's USB cable directly into your computer. Turn the printer on.
- **For Network/Wi-Fi:** Use the printer's built-in screen menu to connect it to your home Wi-Fi network. Ensure your computer is on that same Wi-Fi network.

Step 2: Add the Printer & Install Drivers

1. Open the Start Menu and click Settings (the gear icon).
2. Go to Bluetooth & devices, then click Printers & scanners.
3. Click the Add device button at the top.
4. Wait for Windows to find your printer, then click Add device next to its name.
5. Windows will automatically download and install the correct driver background.

Step 3: Print a Test Page

1. In that same Printers & scanners menu, click on your newly added printer.
2. Click Print test page (usually the first option at the top).
3. Check the printer tray to confirm a page with the Windows logo and diagnostic text successfully prints.

Q3. Simulate a connectivity issue by unplugging the printer's USB cable, then use Windows Device Manager to identify and resolve the 'printer not detected' problem. Hint: Check for error icons and use the 'Scan for hardware changes' option after reconnecting.

❖ Resolve the 'printer not detected' problem

- Unplug the printer USB cable from the computer. This simulates a connectivity problem.
- Open Device Manager by right-clicking the Start button and selecting Device Manager.
- Expand Printers or Print queues and check if the printer is missing or shows a yellow warning icon.
- Reconnect the printer's USB cable securely to the computer.
- In Device Manager, click Action → Scan for hardware changes.
- Windows should detect the connected printer and reinstall/load the required driver.
- If a yellow error icon remains, right-click the printer and select Update driver or Uninstall device, then scan for hardware changes again.
- Open Settings → Bluetooth & devices → Printers & scanners and confirm that the printer is detected.
- Print a test page to verify that the connectivity problem has been resolved.

Q4. A user reports that their webcam video is not displaying in Zoom, but it works in the default camera app. Troubleshoot and resolve the issue so the webcam works in both applications.: Check if another app is using the webcam or if Zoom permissions are enabled.

❖ **Troubleshoot Webcam Not Displaying in Zoom---**

1. Test the webcam in the Camera app Open the Windows Camera app and confirm that the webcam video is working. Since it works there, the webcam and its driver are likely functioning properly.

2. Close other apps using the webcam Close the Camera app and any other applications such as Microsoft Teams, Skype, Google Meet, or Discord that may be using the webcam. A webcam may not work in Zoom if another application has control of it.

3. Check Windows camera permissions Go to Settings → Privacy & security → Camera and make sure:

- Camera access is turned ON.
- Let apps access your camera is turned ON.
- Let desktop apps access your camera is turned ON.

4. Check Zoom camera settings Open Zoom → Settings → Video and check whether the correct webcam is selected under Camera.

5. Test the camera in Zoom Start or join a Zoom meeting and click the arrow next to the Video icon. Select the correct webcam and check whether the video appears.

6. Restart Zoom Completely Close Zoom and open it again. If necessary, restart the computer to release the webcam from any application that may still be using it.

7. Check the webcam driver Open Device Manager → Cameras, right-click the webcam, and select Update driver. Make sure there is no warning/error icon.

8. Verify the solution Open the Camera app again and then test the webcam in Zoom. The webcam should now display video in both applications.